

ClimaX – Carbon Credit Exchange Platform for Sustainable India

- Theme: Sustainable & Green Technology
- Mentor: Dr. Anurag Sewak

Team Members

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Objective

- Build a web-based carbon credit marketplace
- Enable verified carbon credit creation & trading
- Simplify document-based & auditor-based verification
- Ensure traceability using blockchain
- Design sustainable revenue model

Motivation

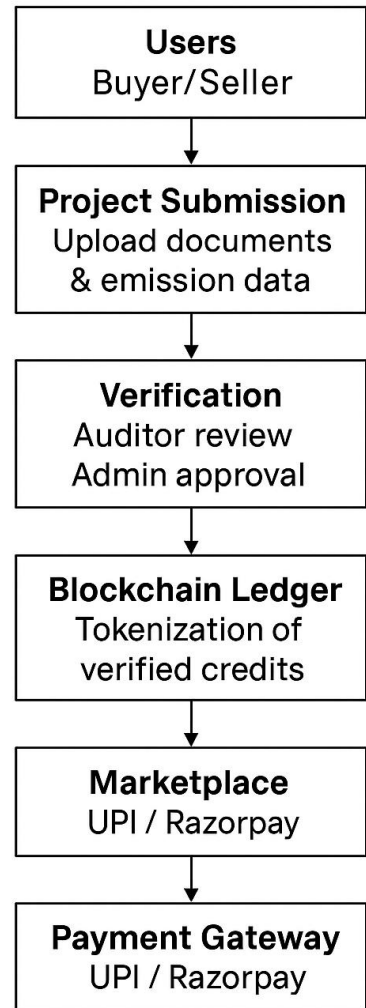
- India targets **Net Zero** by 2070
- Growing need for **transparent carbon trading**
- Lack of accessible digital platform for all participants
- Empower MSMEs, NGOs & individuals to **monetize sustainability**
- Promote India's **green-fintech innovation**

Proposed Model



- Full-stack software-only web app
- Role-based system: Buyer | Seller | Auditor | Admin
- Blockchain for verified credit tokens and transactions
- Real-time dashboards & analytics

Block Diagram



Dashboards & Analytics

Technical Requirements



Frontend:

- React.js / Next.js, Tailwind CSS

Backend:

- Node.js, Express.js

Blockchain:

- Ethereum / Solana / Polygon

Database:

- PostgreSQL + PostGIS, MongoDB

Deployment & Hosting:

- GitHub, Vercel, Render

Progress Till Date

- Project idea finalized and validated
- Complete architecture & data flow designed
- High-Level Design (HLD) and Low-Level Design (LLD) completed
- API documentation prepared for all major modules
- Backend structure and database schema created
- Ready to begin module-wise implementation phase

Timeline

- **Phase 1 (Weeks 1–4):** Core Backend & MVP Development
- **Phase 2 (Weeks 5–8):** Verification & Blockchain Integration
- **Phase 3 (Weeks 9–12):** Marketplace & Admin Dashboard
- **Phase 4 (Weeks 13–16):** Testing, Pilot Run & Feedback Integration

References

- Bureau of Energy Efficiency (BEE) Indian Carbon Market & Registry Guidelines:
<https://beeindia.gov.in/sites/default/files/Detailed%20Procedure%20for%20Compliance%20Procedure%20under%20CCTS.pdf>
- Carbon Credit Trading Scheme (CCTS 2023): <https://icapcarbonaction.com/en/ets/indian-carbon-credit-trading-scheme>
- *State and Trends of International Carbon Markets* (World Bank):
<https://openknowledge.worldbank.org/server/api/core/bitstreams/2eb25e8e-ca16-4649-b637-e5caf88fd625/content>
- Green Credit Programme (MoEFCC) market-based mechanism for environment actions:
<https://spiceroutelegal.com/publications/navigating-carbon-credits-in-india>

Thank you

